

Notes: Goldstein and Pauzner (JF, 2005)

Demand-Deposit Contracts and the Probability of Bank Runs

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1 Big picture

- **Question.** Diamond and Dybvig (1983) show that demand-deposit contracts provide liquidity insurance but also expose banks to self-fulfilling runs. Goldstein and Pauzner ask: once the probability of panic-based runs is endogenously determined, do demand deposits still improve welfare, and how should the short-term promised payment be chosen?
- **Core problem in Diamond–Dybvig.** The original model has a good equilibrium and a run equilibrium, but it does not assign probabilities to them. Therefore, it cannot answer whether banks are desirable ex ante or how the contract affects the probability of runs.
- **Main innovation.** The paper adds stochastic fundamentals and noisy private signals. This converts the multiple-equilibrium Diamond–Dybvig environment into a unique Bayesian equilibrium in which a run occurs if and only if fundamentals fall below a threshold.
- **Panic remains central.** Runs are not purely fundamental in the sense of being individually optimal regardless of others. In the intermediate region, depositors run because they expect others to run. Fundamentals coordinate beliefs rather than mechanically forcing actions.
- **Contract-run feedback.** A higher period-1 promised payment r_1 provides more liquidity insurance, but it also raises the threshold below which runs occur. Thus, more risk sharing makes the bank more fragile.
- **Welfare result.** If the range of fundamentals in which early liquidation is efficient is not too large, the optimal r_1 is above the autarkic liquidation value. Banks are viable even after the endogenous probability of runs is accounted for.
- **Optimal contract result.** The optimal r_1 is below the first-best risk-sharing level c_1^{FB} . The bank deliberately leaves some risk-sharing gains unexploited to reduce run risk.
- **Technical contribution.** The uniqueness proof does not rely on global strategic complementarities. Bank-run payoffs have only one-sided strategic complementarities because the incentive to run peaks when the bank is just close to failing, not when everyone runs.
- **Economic takeaway.** Demand deposits are welfare-improving only after trading off liquidity insurance against endogenous fragility. The contract itself determines the probability of a panic-based run.

2 Incremental contribution of the paper

- **From equilibrium multiplicity to a run probability.** The paper turns Diamond–Dybvig’s binary equilibrium problem into a model with a unique threshold equilibrium and a computable ex ante run probability.
- **Endogenizing fragility.** Unlike models where the probability of a bank run is exogenous, here the probability of a run is determined by the deposit contract. A more generous short-term payment increases fragility.
- **Maintaining panic-based runs.** Some global-games models eliminate panic by making runs purely fundamental. This paper keeps the panic mechanism: depositors run in the intermediate region only because they expect other depositors to run.

- **A new uniqueness argument.** The paper extends global-games logic to settings with only one-sided strategic complementarities, which is essential for standard bank-run payoffs.
- **Contract design under endogenous crisis risk.** The optimal demand-deposit contract internalizes the effect of liquidity insurance on run probabilities. The optimal short-term payment is neither autarky nor first-best risk sharing.
- **Policy relevance.** Once run probabilities are endogenous, one can evaluate deposit insurance, suspension of convertibility, lender-of-last-resort policies, and other interventions by asking how they affect the contract and the run threshold.
- **Broader finance relevance.** The proof technique applies naturally to debt rollover, IPO coordination, financial-market runs, and other settings in which strategic complementarities are strong only on one side of the state space.

3 Model environment

3.1 Agents, timing, and preferences

There are three periods, 0, 1, and 2, and a continuum of agents indexed by $[0, 1]$. Each agent is endowed with one unit of the good at date 0. Consumption occurs only at date 1 or date 2.

Each agent can be one of two types:

- with probability λ , the agent is **impatient** and can consume only at date 1;
- with probability $1 - \lambda$, the agent is **patient** and can consume at date 1 or date 2.

Types are privately learned at the beginning of period 1. Utility is $u(c_1)$ for impatient agents and $u(c_1 + c_2)$ for patient agents. The utility function is increasing, twice differentiable, and sufficiently risk averse: for any $c > 1$, relative risk aversion is greater than 1.

3.2 Technology and fundamentals

The economy has a long-term technology. One unit invested at date 0 can be liquidated at date 1 for one unit. If held to date 2, it yields R with probability $p(\theta)$ and 0 otherwise. The state θ is uniformly distributed on $[0, 1]$, and $p(\theta)$ is increasing.

The assumption

$$E_\theta[p(\theta)]u(R) > u(1)$$

means that, for patient agents, the long-term asset is ex ante superior to early liquidation.

3.3 First-best risk sharing

If types were observable, a planner would choose the period-1 consumption of impatient agents, c_1 , to solve

$$\max_{c_1} \lambda u(c_1) + (1 - \lambda) E_\theta \left[p(\theta) u \left(\frac{1 - \lambda c_1}{1 - \lambda} R \right) \right].$$

The first-best short-term consumption c_1^{FB} satisfies

$$u'(c_1^{FB}) = Ru' \left(\frac{1 - \lambda c_1^{FB}}{1 - \lambda} R \right) E_\theta[p(\theta)].$$

Because the marginal gain from transferring resources to impatient agents is high at $c_1 = 1$, the first-best has $c_1^{FB} > 1$. This is the liquidity-insurance motive for demand deposits.

4 Demand-deposit contract

4.1 Contract form

A bank offers a demand-deposit contract. Each agent deposits her endowment at date 0. At date 1, an agent who withdraws is promised a fixed payment $r_1 > 1$. An agent who waits until date 2 receives the proceeds of the remaining investment divided among the remaining depositors.

The bank follows a sequential-service constraint in period 1: it pays r_1 until resources are exhausted. This is why a run can make some early withdrawers receive zero.

4.2 Payments as a function of withdrawals

Let n denote the total fraction of agents who demand early withdrawal. If $n < 1/r_1$, the bank has enough resources to pay all early withdrawers. If $n \geq 1/r_1$, the bank exhausts its liquidated resources before serving all withdrawers.

The payment structure is summarized in Table I from the paper. When $n < 1/r_1$, early withdrawers get r_1 and late withdrawers share the remaining long-term asset. When $n \geq 1/r_1$, early withdrawers are served only with probability $1/(nr_1)$ and late withdrawers get zero.

4.3 Diamond–Dybvig equilibria

If the bank sets $r_1 = c_1^{FB}$ and only impatient depositors withdraw at date 1, the first-best allocation can be implemented. But there is also a run equilibrium: if all agents expect everyone else to withdraw, then withdrawing is optimal for patient agents as well. The Diamond–Dybvig model therefore shows both the benefit and fragility of demand deposits but does not determine which equilibrium occurs.

5 Private signals and unique equilibrium

5.1 Information structure

At the beginning of period 1, the state θ is realized but not publicly observed. Each agent i receives a private signal

$$\theta_i = \theta + \varepsilon_i,$$

where ε_i is uniformly distributed on $[-\varepsilon, \varepsilon]$ and independent across agents.

A higher signal does two things:

- it makes the agent more optimistic about the long-term return;
- it makes the agent believe other agents likely received high signals and therefore are less likely to run.

5.2 Dominance regions

The model assumes two extreme regions:

- **Lower dominance region.** Fundamentals are so bad that a patient agent wants to withdraw even if all other patient agents wait.
- **Upper dominance region.** Fundamentals are so good that a patient agent wants to wait even if many others withdraw.

These extreme regions are small but important. They anchor the equilibrium and generate a contagion of reasoning that pins down behavior in the intermediate region.

5.3 Threshold equilibrium

Theorem 1 states that the model has a unique equilibrium in which patient agents use a cutoff strategy:

$$\text{patient agent runs} \iff \theta_i < \theta^*(r_1).$$

Given the cutoff $\theta^*(r_1)$, the fraction of agents who withdraw at state θ is

$$n(\theta, \theta^*) = \begin{cases} 1, & \theta < \theta^*(r_1) - \varepsilon, \\ \lambda + (1 - \lambda) \left(\frac{1}{2} + \frac{\theta^*(r_1) - \theta}{2\varepsilon} \right), & \theta^*(r_1) - \varepsilon \leq \theta \leq \theta^*(r_1) + \varepsilon, \\ \lambda, & \theta > \theta^*(r_1) + \varepsilon. \end{cases}$$

Thus, the realized fundamentals determine the measure of agents running, and hence determine whether a run occurs. However, in the intermediate region the run is still panic-based because each patient depositor runs only because she expects other depositors to run.

6 Run probability and contract design

6.1 Run threshold as a function of r_1

When noise becomes small, the equilibrium threshold is characterized by an indifference condition for the marginal patient depositor. The paper derives

$$\lim_{\varepsilon \rightarrow 0} \theta^*(r_1) = p^{-1} \left(\frac{u(r_1) (1 - \lambda r_1 + \ln r_1)}{r_1 \int_{\lambda}^{1/r_1} u \left(\frac{1 - nr_1}{1 - n} R \right) dn} \right).$$

The key comparative static is Theorem 2:

$$\theta^*(r_1) \text{ is increasing in } r_1.$$

A higher promised short-term payment makes early withdrawal more attractive and makes late withdrawal less attractive. It also makes each depositor expect more withdrawals by others. Therefore, the bank becomes fragile in a larger set of fundamentals.

6.2 Ex ante welfare

Expected welfare under contract r_1 integrates over run and no-run states:

$$EU(r_1) = \int_0^{\theta^*(r_1)} \frac{1}{r_1} u(r_1) d\theta + \int_{\theta^*(r_1)}^1 \left[\lambda u(r_1) + (1 - \lambda) p(\theta) u\left(\frac{1 - \lambda r_1}{1 - \lambda} R\right) \right] d\theta.$$

The first term corresponds to run states, in which all investment is liquidated and agents receive r_1 only with probability $1/r_1$. The second term corresponds to no-run states, in which impatient agents receive r_1 and patient agents receive the long-term payoff.

6.3 Viability of banks

Theorem 3 states that if the lower dominance region at $r_1 = 1$ is not too large, the optimal short-term payment satisfies

$$r_1^* > 1.$$

Interpretation: demand deposits are welfare-improving even after accounting for panic runs, provided early liquidation is efficient only for a limited range of very bad fundamentals. The first-order gain from liquidity insurance dominates the small marginal crisis cost near $r_1 = 1$.

6.4 Optimal contract below first-best risk sharing

Theorem 4 states

$$r_1^* < c_1^{FB}.$$

The bank does not implement the first-best risk-sharing level because raising r_1 has two destabilizing costs:

- it increases the probability of a run by increasing $\theta^*(r_1)$;
- it increases the cost of a run because, under sequential service, more promised payment to each early withdrawer means fewer agents can be served.

7 Tables and figures

7.1 Table I and Figure 1: Demand-deposit payments and dominance regions

Read:

- Table I lays out the ex post payoffs under the demand-deposit contract. If $n < 1/r_1$, the bank can pay all early withdrawers. If $n \geq 1/r_1$, it exhausts resources and late withdrawers receive zero.

- The table makes clear why the sequential-service constraint is essential: in a run, some early withdrawers may be paid while others receive nothing.
- Figure 1 shows the lower and upper dominance regions. In the lower dominance region, all patient agents surely run; in the upper dominance region, no patient agents run.
- The intermediate region is where beliefs matter. The dominance regions create bounds on behavior, and noisy signals propagate these bounds inward until a unique threshold equilibrium is pinned down.

Economic message: Demand deposits create risk sharing but also a discontinuous payoff structure around the bankruptcy point. Dominance regions are the anchors that select a unique threshold equilibrium.

Table I
Ex Post Payments to Agents

Withdrawal in Period	$n < 1/r_1$	$n \geq 1/r_1$
1	r_1	$\begin{cases} r_1 & \text{with probability } \frac{1}{nr_1} \\ 0 & \text{with probability } 1 - \frac{1}{nr_1} \end{cases}$
2	$\begin{cases} \frac{(1-nr_1)}{1-n} R & \text{with probability } p(\theta) \\ 0 & \text{with probability } 1 - p(\theta) \end{cases}$	0

Table I: Ex post payments to agents.

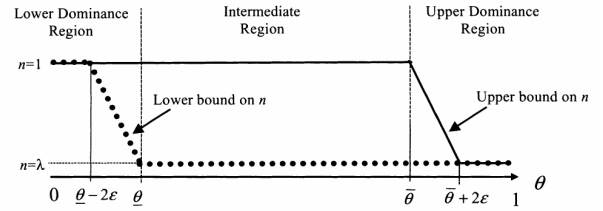


Figure 1: Direct implications of the dominance regions on agents' behavior.

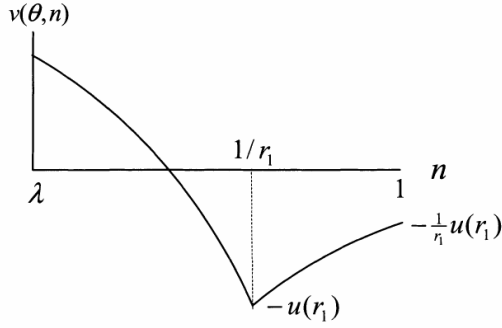
Figure 1: Direct implications of dominance regions.

7.2 Figures 2 and 3: One-sided strategic complementarities and threshold uniqueness

Read:

- Figure 2 plots a patient agent's net incentive to wait until period 2 rather than withdraw at period 1, $v(\theta, n)$, as a function of the share of withdrawers n .
- The incentive to run is strongest not when everyone runs, but when n reaches the insolvency point $1/r_1$. Once the bank is already insolvent, additional runners reduce the chance of being served and the run incentive weakens.
- Therefore, the model does not satisfy global strategic complementarities.
- Figure 3 illustrates the threshold logic: at θ^* the marginal depositor is indifferent; signals below θ^* shift probability mass toward negative values of v , and signals above θ^* shift mass toward positive values of v .

Economic message: The uniqueness proof is not a standard global-games proof. It relies on a single-crossing structure and one-sided strategic complementarities, which are more appropriate for bank-run payoffs.



2. The net incentive to withdraw in period 2 versus period 1.

Figure 2: Net incentive to withdraw in period 2 versus period 1.

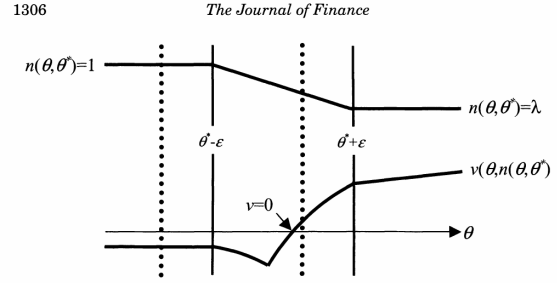


Figure 3. Functions $n(\theta, \theta^*)$ and $v(\theta, n(\theta, \theta^*))$.

Figure 3: Functions $n(\theta, \theta^*)$ and $v(\theta, n(\theta, \theta^*))$.

7.3 Figure A1: Why nonthreshold equilibria are ruled out

Read:

- Figure A1 appears in the appendix proof of Theorem 1.
- It depicts a counterfactual nonthreshold equilibrium with two signals θ_A and θ_B where the depositor is indifferent.
- Between these points, the proof would require incompatible signs of the payoff differential Δ .
- The figure supports the argument that any equilibrium must be a threshold equilibrium.

Economic message: The appendix figure is technical, but it reinforces the main result: once noisy private information and dominance regions are introduced, the equilibrium cannot have arbitrary switching patterns; it must be a cutoff rule.

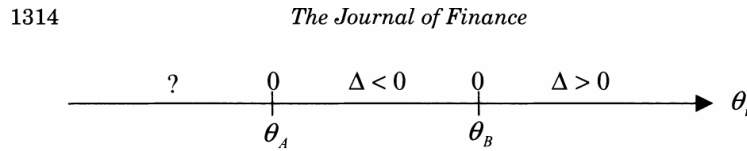


Figure A1: Function Δ in a counterfactual nonthreshold equilibrium.

8 Detailed interpretation of the four main theorems

8.1 Theorem 1: unique threshold equilibrium

Theorem 1 solves the Diamond–Dybvig equilibrium-selection problem. Once agents receive noisy signals about fundamentals, there is a unique cutoff signal. Patient agents withdraw below the cutoff and wait above it. The cutoff transforms a multiple-equilibrium bank-run model into a model with a well-defined run probability.

The result is economically important because it preserves panic-based runs. In the intermediate region, a patient depositor would not run if she believed others would not run, but she runs because her signal leads her to believe others are likely to run.

8.2 Theorem 2: liquidity provision increases fragility

Theorem 2 shows that $\theta^*(r_1)$ is increasing in r_1 . A larger promised short-term payment increases risk sharing but also makes early withdrawal more attractive and reduces the continuation value of waiting. Therefore, a bank that provides more liquidity insurance becomes more fragile.

This is the central contract-fragility feedback. The demand-deposit contract is not a passive object; it changes the probability of the bad event it exposes the bank to.

8.3 Theorem 3: banks are viable

Theorem 3 provides a condition under which the optimal r_1 is above 1. In other words, the bank should provide liquidity insurance, not simply replicate autarky.

The key intuition is marginal. When r_1 is raised slightly above 1, the benefit from risk sharing is first order because impatient agents receive valuable liquidity. The cost from a marginally higher run probability is small when the extra run states are close to the efficient-liquidation boundary.

8.4 Theorem 4: the optimal contract is conservative

Theorem 4 states that $r_1^* < c_1^{FB}$. The first-best planner ignores panic-based runs because types are observable and there is no run problem. The bank, by contrast, must internalize the fact that a larger r_1 makes the system more fragile. Hence the optimal demand-deposit contract provides partial but not full risk sharing.

9 Short summary

- The paper extends Diamond–Dybvig by adding stochastic fundamentals and noisy private signals.
- This creates a unique Bayesian threshold equilibrium: a bank run occurs iff fundamentals are below a critical threshold.
- Runs in the intermediate region are panic-based because agents run due to expectations about others' withdrawals.
- The model computes the ex ante probability of panic-based bank runs.
- A higher promised short-term payment r_1 increases liquidity insurance but also increases the run threshold and the probability of a run.
- Demand deposits are welfare-improving if the range of fundamentals where early liquidation is efficient is not too large.
- The optimal r_1 is above autarky but below the first-best risk-sharing level.

- The model gives a framework for analyzing deposit insurance, suspension of convertibility, and lender-of-last-resort policies because run probabilities are endogenous.

10 Conclusion

10.1 Core conclusion

Goldstein and Pauzner show that demand-deposit contracts should be evaluated by jointly considering their liquidity-insurance benefit and their endogenous fragility cost. The paper resolves the equilibrium-selection problem in Diamond–Dybvig by using noisy private signals, yielding a unique threshold equilibrium and a computable run probability.

10.2 Economic intuition

Demand deposits are valuable because they transfer resources to impatient agents, who urgently need liquidity. But the same short-term promise that creates liquidity insurance also makes patient agents more willing to run when they fear others will run. The optimal contract balances these opposing forces.

10.3 Incremental contribution revisited

The paper’s incremental contribution is not merely adding fundamentals to bank runs. It shows how fundamentals coordinate expectations in a way that keeps the panic interpretation alive. The result is a model in which bank runs are predictable in probability, contract-dependent, and still self-fulfilling in the relevant region.

10.4 Implications for banking policy

The model suggests that policies should not be evaluated only by whether they remove panic. They must also account for how contracts respond. Deposit insurance, for example, may reduce run incentives but may also induce banks to set overly generous short-term promises, raising moral hazard and fragility elsewhere.

10.5 Limitations

The model is stylized: it abstracts from interbank networks, asset fire sales, bank capital, heterogeneous banks, regulatory constraints, and macro feedback. It also focuses on one representative bank and a highly structured signal environment. Nonetheless, the mechanism is general: contracts that create liquidity can also create run risk.

10.6 Takeaway

The paper’s lasting insight is that the probability of a panic run is not exogenous. It is a function of the financial contract. More liquidity insurance means more fragility, so optimal banking contracts

deliberately provide less than first-best risk sharing.